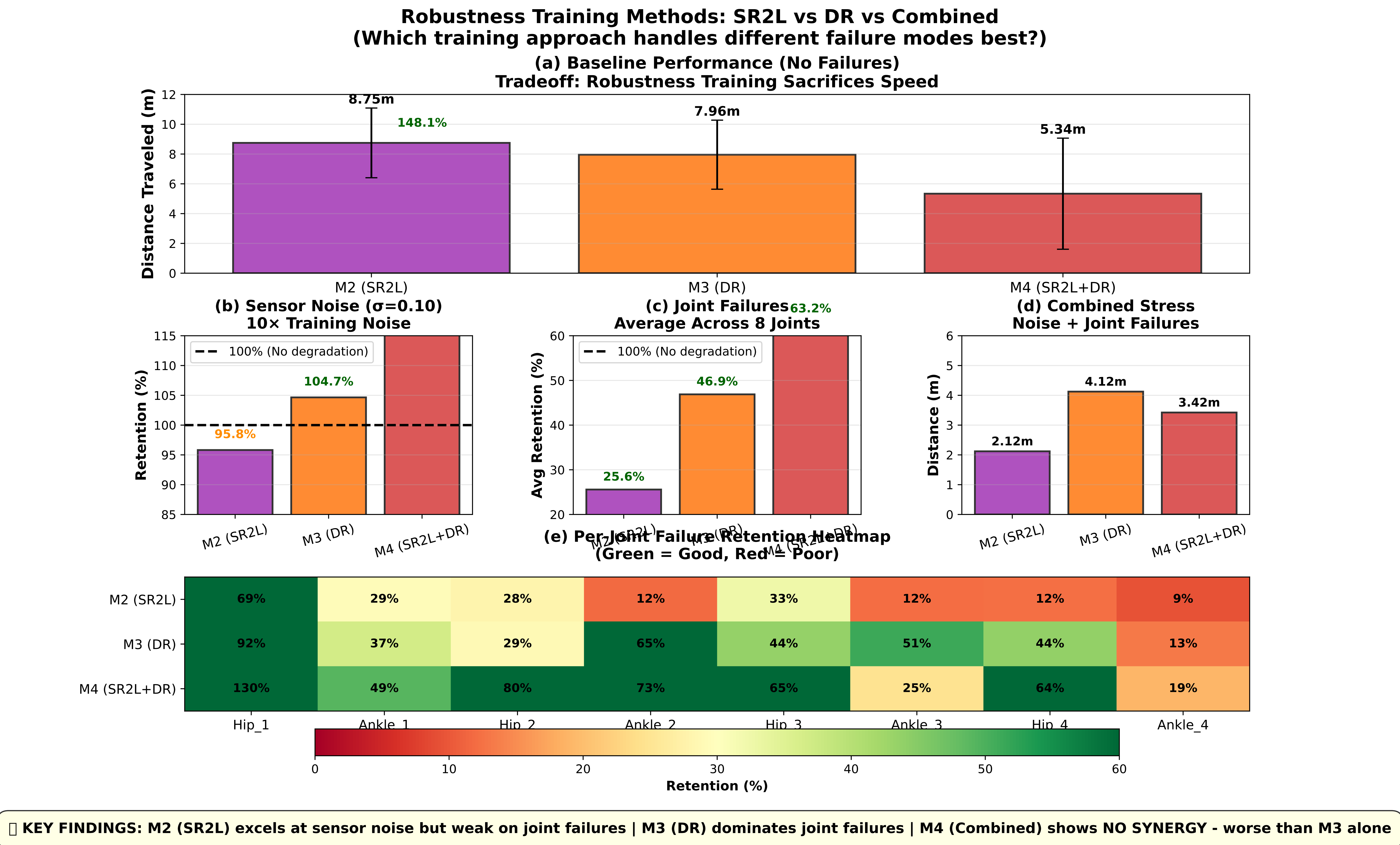
Robustness Training Methods: SR2L vs DR vs Combined
(Which training approach handles different failure modes best?)
(a) Baseline Performance (No Failures)
Tradeoff: Robustness Training Sacrifices Speed
Distance Traveled (m)
0
2
4
6
8
10
12
8.75m
148.1%
7.96m
5.34m
M2 (SR2L)
M3 (DR)
M4 (SR2L+DR)
(b) Sensor Noise (σ=0.10)
10× Training Noise
Retention (%)
85
90
95
100
105
110
115
100% (No degradation)
95.8%
104.7%
M2 (SR2L)
M3 (DR)
M4 (SR2L+DR)
(c) Joint Failures
Average Across 8 Joints
Avg Retention (%)
20
30
40
50
60
100% (No degradation)
25.6%
46.9%
63.2%
M2 (SR2L)
M3 (DR)
M4 (SR2L+DR)
(d) Combined Stress
Noise + Joint Failures
Distance (m)
0
1
2
3
4
5
6
2.12m
4.12m
3.42m
M2 (SR2L)
M3 (DR)
M4 (SR2L+DR)
(e) Per-Joint Failure Retention Heatmap
(Green = Good, Red = Poor)
M2 (SR2L)
M3 (DR)
M4 (SR2L+DR)
69%
29%
28%
12%
33%
12%
12%
9%
92%
37%
29%
65%
44%
51%
44%
13%
130%
49%
80%
73%
65%
25%
64%
19%
Hip_1
Ankle_1
Hip_2
Ankle_2
Hip_3
Ankle_3
Hip_4
Ankle_4
0
10
20
30
40
50
60
Retention (%)
KEY FINDINGS: M2 (SR2L) excels at sensor noise but weak on joint failures | M3 (DR) dominates joint failures | M4 (Combined) shows NO SYNERGY - worse than M3 alone